Cookie Cats: A/B Testing & Retention Analysis

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1. Defining The Problem

Cookie Cats is a mobile puzzle game featuring progression gates that temporarily restrict player advancement. The experiment evaluates whether moving the first progression gate from Level 30 (control group) to Level 40 (treatment group) improves player engagement and retention. This analysis examines the results of an A/B experiment to evaluate the impact of gate placement on player retention. The question we would like to answer is the following: Should the first progression gate remain at Level 30 or be moved to Level 40?

2. Methodology

The analysis uses a publicly available A/B testing dataset containing **90,189 players**, assigned to two experimental groups, according to in which level the gate was introduced. For 44,700 players, the gate was introduced at Level 30, whereas for 45,489 players, the gate was introduced at Level 40.

In order to understand the effects of this test, day1 (D1) and day7 (D7) retention are analyzed. This was done using two-proportion z-tests with a 5% significance threshold. Gameplay rounds were analyzed using descriptive statistics.

For this analysis, D7 retention was treated as the principal decision metric.

3. Key Findings

Metric	Level 30	Level 40
D1 Retention	44.82%	44.23%
D7 Retention	19.02%	18.20%
Mean Gameplay Rounds	52.46	51.30
Median Gameplay Rounds	17	16

Statistical results

For D1 retention, the difference observed was not statistically significant, with the p-value being 0.07441. This was not the case in D7 retention. When the gate was placed in Level 40, D7 showed a statistically significant decrease of approximately 0.82 percentage points. The p-value was measured to be 0.00155. The 95% confidence interval for the retention difference (Gate 40 minus Gate 30) ranged from -1.33 to -0.31 percentage points.

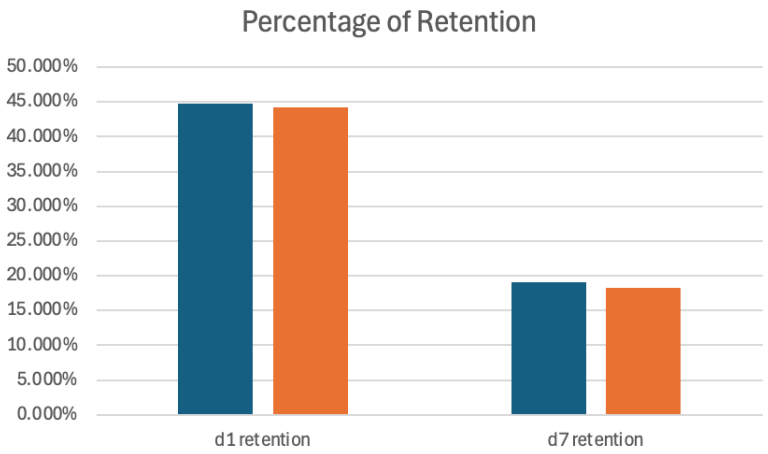


Figure1: D1 and D7 Retention. Blue: level 30. Orange: Level 40.

As shown in Figure 1, the retentions are similar in D1, but when we look at D7 retention, the case in which the gate is at level 30, outperforms the case in which the gate is at level 40.

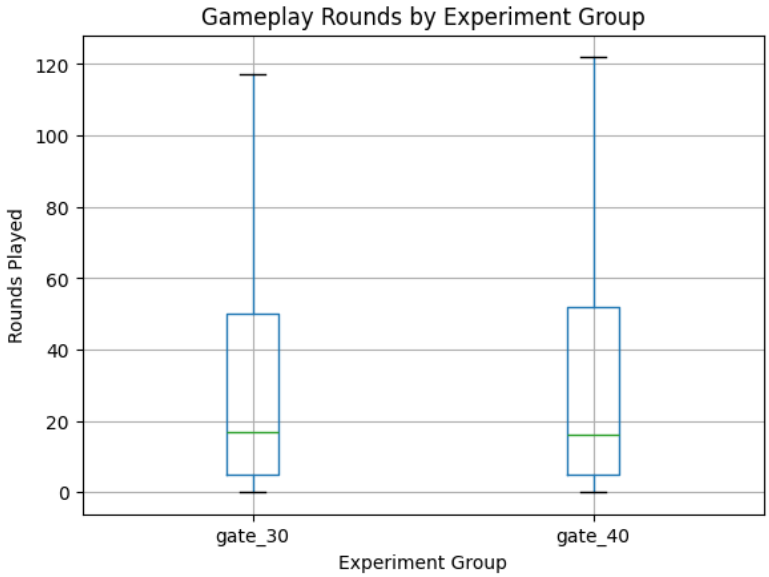


Figure 2: Distribution of Gameplay Rounds by Experiment Group.

Gameplay rounds showed broadly similar distributions across the two groups. Gate 30 had marginally higher mean and median values, while Gate 40 showed a slightly higher upper quartile in the boxplot. Extreme observations were hidden from the visualization but retained in the calculations. The descriptive differences do not establish a statistically significant effect on gameplay engagement.

4. Experiment Validity and Limitations

Experiment Validity

The experiment included 44,700 players in Gate 30 and 45,489 players in Gate 40. A chi-square goodness-of-fit test was conducted to find out that this difference would be unusual if players were expected to be divided equally between the two groups. The value of p was found to be 0.00861. However, the original intended group allocation is unknown. Therefore, this result does not prove that the experiment was incorrectly conducted, but it highlights a potential issue that should be investigated.

Data Limitations

The dataset provides information about player retention and gameplay rounds but does not include revenue, in-app purchases, session duration, or progression-gate interactions. As a result, the analysis can compare retention performance but cannot determine how the gate change affected monetization or explain exactly why players returned or stopped playing.

5. Conclusion and Recommendations

Based on the available evidence, I would suggest against rolling out the Level 40 gate. I would retain the gate at Level 30 provisionally while investigating experiment validity. A 0.82 percentage point drop in D7 retention translates to roughly 8,200 fewer D7 active users per 1 million users. Assuming comparable player populations, a 0.82 percentage-point decline in D7 retention would correspond to approximately 8,200 fewer Day-7 returning players per 1 million new players.

The observed D7 retention results favor Gate 30, with no statistically significant D1 retention improvement from moving the gate to Level 40. However, the potential sample-ratio mismatch requires investigation before the findings are treated as definitive causal evidence. Monetization outcomes must also be evaluated before making a final business decision.

After validating the assignment process, a follow-up experiment could compare progression gates at Levels 30, 35, and 40. The next study should track D1/D7 retention, gate reach and completion events, session behavior, payer conversion, and average revenue per daily active user (ARPDAU). This would help determine whether an alternative gate location can improve retention while maintaining or increasing monetization.

Data source: Mobile Games: A/B Testing — Kaggle

Tools used: Microsoft Excel, Python, pandas, SciPy, statsmodels, Matplotlib.